MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF AERONAUTICAL AND AUTOMOBILE ENGINEERING			
END SEMESTER EXAMINATION		**NOVEMBER, 2024**	
Level	**B.Tech**	**Curriculum**	**2018**
Program Name	**Automobile Engineering**	**Course Type**	**PE**
Course Name	**Automotive pollution and control**	**Course Code**	**AAE4074**
Semester	**VII**	**Max Marks**	**50**
Date	**25/11/2024**	**Duration**	**3 hrs**

Instructions:

- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		IET*		BL
			CO	PO	CLO	AHEP4 LO	
1A	Illustrate with a relevant diagram, how different modes of operation are executed in a TOYOTA lean burn engine.	**04**	3	1,2,3	**3**	**1,12**	**3**
1B	With a neat sketch, discuss the S I engine set up required to operate using natural gas. What are the advantages of natural gas when used for engine operation?	**03**	4	4,5	**4**	**1,4,12,13**	**2**
1C	With a neat sketch, explain the working principle of an air diverter valve used in air injection system.	**03**	1	10	**1**	**1,4,18**	**2**
2A	Discuss the effects of the following factors on smoke intensity exhausted from diesel engines (i) Injection system (ii) Engine load (iii) Fuel properties (iv) Engine speed	**04**	2	1,2,4	**2**	**1,4,12**	**3**
2B	With a neat sketch, explain the working of P.C.I Volkswagon stratified charge engine. What are its merits?	**03**	3	1,2,5	**3**	**1,12**	**2**

2C	Illustrate with a relevant sketch, the effectiveness of positive crankcase ventilation system in minimizing the engine emissions	**03**	1	1,4,5	**1**	**2,12,13**	**3**
3A	Discuss the effects of the following factors on performance of dual fuel engines. (i) Primary fuel (ii) Pilot fuel quantity (iii) Cetane number of pilot fuel (iv) Inlet temperature of primary fuel	**04**	4	1,210	**3**	**1,4,12**	**3**
3B	With a relevant diagram, illustrate the working principle of any one detector system used in gas chromatography	**03**	2	10,12	**2**	**2,12**	**3**
3C	Discuss the method used for smoke intensity quantification in Hartridge smoke meter	**03**	2	3,5	**2**	**1,12**	**2**
4A	Discuss the effectiveness of the following techniques for dealing with pollution in automotive engines (i) Water injection (ii) Barium salt additives in diesel (iii) Ammonia injection (iv) After burner	**04**	5	1,2,5	**5**	**1,2,12,13**	**3**
4B	Show the experimental set up for exhaust gas analysis using Orsat apparatus and mention all the features and reagents used	**03**	2	1,10	**2**	**12,13**	**2**
4C	Discuss the mechanics involved in the smog formation. What are the effects of smog on the universe?	**03**	1	12	**1**	**1,2,4**	**2**
5A	If octane is used for combustion in an engine with an excess air supply of 20% compared to stoichiometric air. Write the combustion equations for stoichiometric air and actual air supplied. Determine the Fuel air ratio, Air fuel ratio and equivalence ratio	**04**	2	1,2,3	**2**	**1,2**	**4**
5B	List any four causes for the formation of oxides of Nitrogen. What are the effects of its exposure on human beings? Suggest any four techniques for minimizing their emission from engines	**03**	5	12	**5**	**1,4,13**	**3**
5C	With a block diagram, illustrate the working of a dilution tunnel. Which of the pollutants is assessed using this technique?	**03**	5	10	**5**	**12,13**	**3**

MARK DISTRIBUTION (CO-wise & BTL-wise)

CO / CLO	Total Marks
1	***9***
2	***17***
3	***7***
4	***7***
5	***10***

BL	Total Marks
2	18
3	28
4	04
5	